An Overview of Electrolytes and Acid-Base disorders

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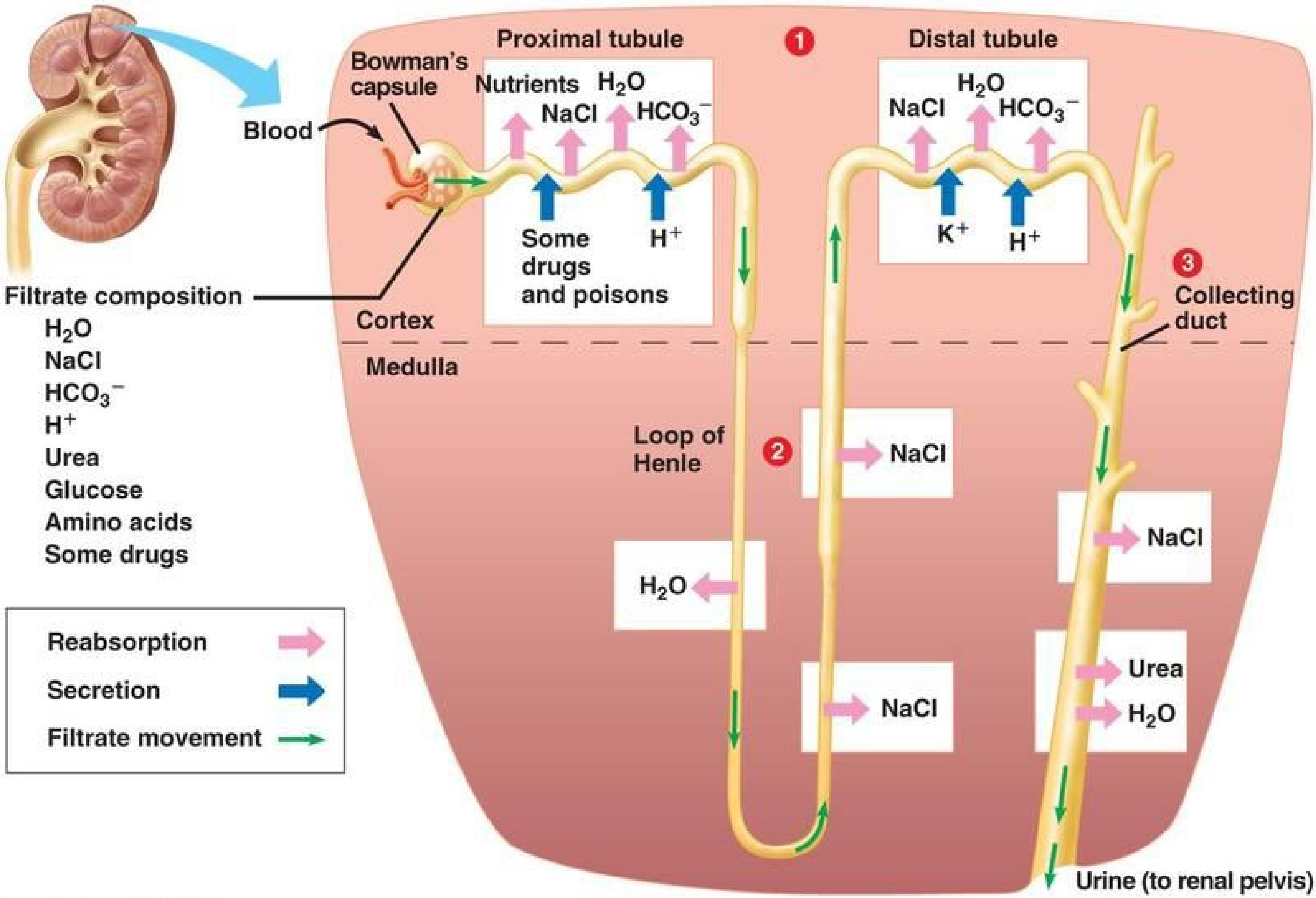
Objectives

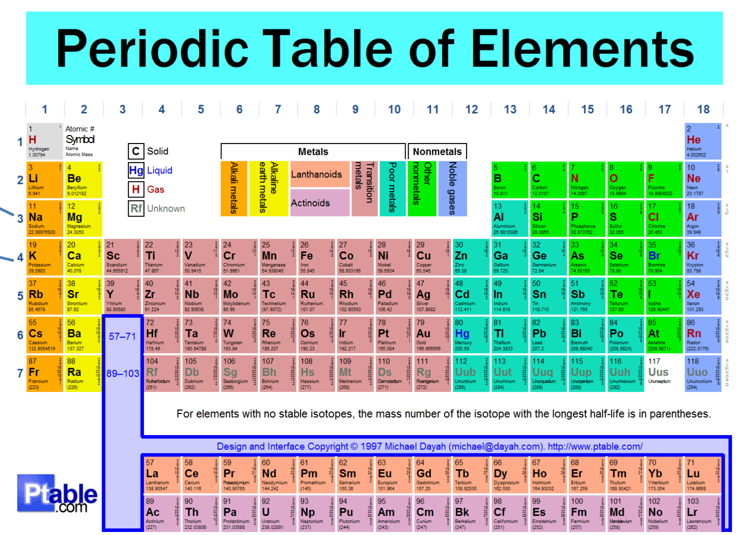
- To understand the mechanism in different types of acid-base and electrolyte disturbances
- To develop a sound approach to investigate and diagnose an electrolyte/acid-base disorder
- To appreciate the general principles for the management of electrolytes/acid-base disturbances



Proximal tubule
1
Distal tubule
Bowman's capsule
Nutrients
H_2O
NaCl
HCO_3^-
Blood
NaCl
H_2O
HCO_3^-
Some drugs and poisons
H^+
K^+
H^+
3
Collecting duct
Filtrate composition
H_2O
NaCl
HCO_3^-
H^+
Urea
Glucose
Amino acids
Some drugs
Cortex
Medulla
Loop of Henle
2
NaCl
NaCl
H_2O
Reabsorption
Secretion
Filtrate movement
NaCl
Urea
H_2O
Urine (to renal pelvis)

Common acid-base and electrolytes problems:

- Acidosis: metabolic/respiratory
- Alkalosis: metabolic/respiratory
- Na (Hypo/Hyper)
- K (Hypo/Hyper)



Acid-base disorders

Acidaemia – $[H^+]$ > normal;

Alkalaemia - $[H^+]$ < normal

Acidosis – a process leading to increase in plasma $[H^+]$

Alkalosis – a process leading to removal of plasma $[H^+]$

Normal pH: 7.4 (7.35-7.45) i.e. pH 7.4 = $[H^+]$ 40 nmol/l; Lethal: pH < 7.1, or > 7.7

Compensation mechanisms

- Lungs – immediate
- Kidneys – takes several days

When well compensated → no acidaemia or alkalaemia despite an underlying acidosis / alkalosis process

Compensatory mechanisms

- Primary metabolic acidosis ($\downarrow HCO_3$)
  - Stimulate resp center $\rightarrow$ $\downarrow pCO_2$ $\rightarrow$ compensatory respiratory alkalosis

- Primary metabolic alkalosis ($\uparrow HCO_3$)
  - Suppress resp center $\rightarrow$ $\uparrow pCO_2$ $\rightarrow$ compensatory respiratory acidosis

- Primary respiratory acidosis ($\uparrow pCO_2$)
  - Kidney compensates by conserving HCO_3 $\rightarrow$ $\uparrow HCO_3$ $\rightarrow$ compensatory metabolic alkalosis

- Primary respiratory alkalosis ($\downarrow pCO_2$)
  - Kidney compensates by excreting HCO_3 $\rightarrow$ $\downarrow HCO_3$ $\rightarrow$ compensatory metabolic acidosis

Metabolic Acidosis

- Characterized by ↓ HCO_3^-
  - Normal [HCO_3^-]: 22-28 mmol/L
  - <22 mmol/L

- Compensated with hyperventilation
  - Reduced pCO_2

Diagnosis of metabolic acidosis

1. Ensure this is Metabolic Acidosis

2. Determine the **Anion Gap** (AG): high AG or normal AG

3. If normal AG:

   Determine Urine AG [i.e. Urine Na + urine K –urine Cl] (Advanced)

   NaHCO3 infusion or Acid loading test: proximal vs. distal RTA (Advanced)

4. Look for any **Osmolar Gap** (OG): Measured osmol – calculated osmol

5. Any mixed acid/base disorder (ΔAG vs. ΔHCO3) ?

   This is too advanced for MBBS

Anion Gap

Total plasma cations = anions

Plasma cations: Na, K, Ca, Mg

- Only Na is present in significant level and may have great variations

Measured anions: Cl, HCO_3^-

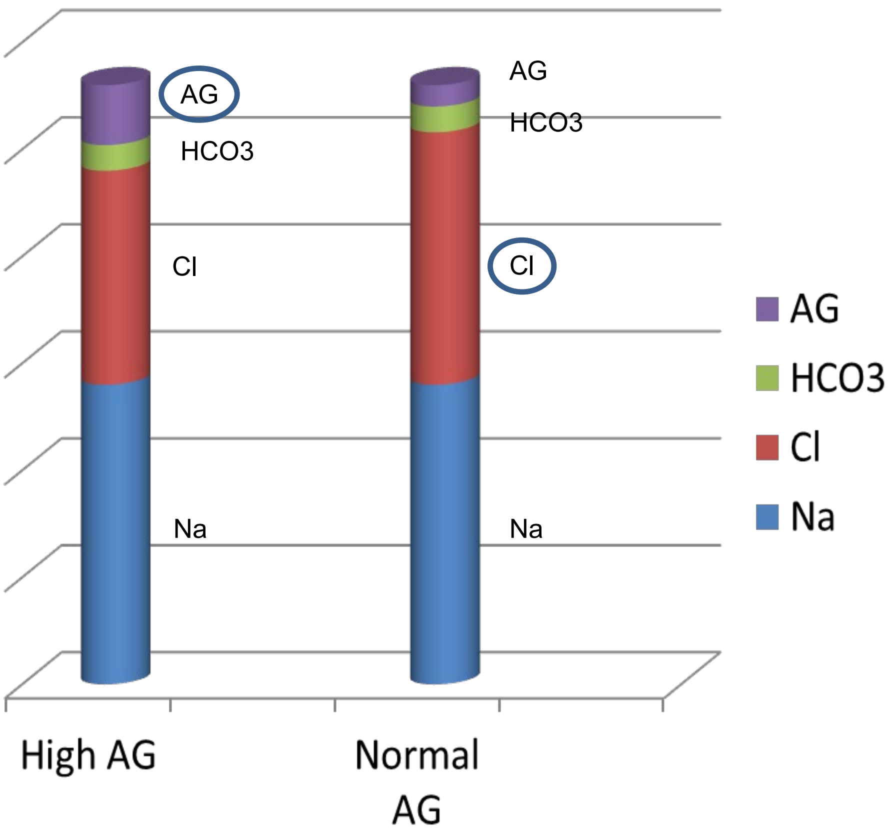
- Measured cations – measured anion = unmeasured anion (i.e. AG)

  Anion gap $= Na - Cl - HCO_3^-$

  e.g. 140 – 105 – 25 = 10

Normal anion gap ~ 8 – 14

Anion Gap in metabolic acidosis



- Similar Na levels
- Both have ↓ HCO_3
- ↑ Cl in Normal AG acidosis
- AG = unmeasured anions

↑ Anion Gap

- ↑ AG → ↑ unmeasured anion in blood
  - DKA , alcoholic ketoacidosis
  - Lactic acid – lactic acidosis
  - Ingestion – salicylate, formic acid (methanol), glycosylate (ethylene glycol)
  - Renal failure – sulfate, phosphate, hippurate and others
  - Rhabdomyolysis – release of organic acids
  - Remarks: altered AG in paraproteinemia (e.g. ↓ in IgG gammopathy; ↑ in IgA gammopathy)

Normal AG Acidosis

Characterized by **↑ [Cl^-]**

Loss of HCO_3^- , with compensatory ↑ Cl^-

- GI loss – diarrhea
- Renal loss – proximal renal tubular acidosis (RTA),carbonic anhydrase inhibitor

Failure to excrete H+

- Distal RTA, type IV RTA

Ingestion of excessive Cl^- - NH_4Cl

Increased reabsorption of Cl^- - ureterosigmoidostomy

Osmolar Gap

- Calculated plasma osmolality
  = 2x[Na^+] + urea + glucose
  (cation = anion, other cations ~ bounded Na+)

- Osmol gap = Measured osmolality - calculated osmolality

- ↑Osmol Gap = presence of an unmeasured osmotically active substance
  - (e.g. ingestion of alcohol related compounds)

L-Lactic acidosis

- Overproduction of L-lactate – O_2 deficiency (type A)
  - Circulatory problem e.g. hypotension
  - Respiratory problem → hypoxia
  - Haemoglobin problem e.g. CO poisoning
  - Increased metabolic demand e.g. grand mal seizure, severe exercise

- Reduced metabolism of L-lactate without hypoxaemia (type B)
  - Liver problem
  - Alcoholism
  - thiamine deficiency
  - Phenormin, metformin

L-lactic acidosis

Rate of production up to 72 mmol/min with total hypoxia in type A (hypoxia)

Diagnosis:

- **high AG metabolic acidosis**
- Measure plasma lactate level
  - Normal: <2 mmol/L

Treatment (type A L-lactic acidosis):

- Most effective: improve O_2 delivery to tissue
  - Correct hypotension, hypoxaemia
- **$NaHCO_3$ therapy ineffective** unless production of lactate controlled
  - Its use may buy time for life saving
  - The Na^+ load limits its massive use
- Haemodialysis with bicarbonate dialysis

Management of metab acidosis

Determine cause of acidosis and treat the underlying cause:

- Some causes have independent threat to life e.g. methanol poisoning

- There maybe specific treatment for certain causes, e.g. methanol poisoning

Correction of HCO_3^- by $NaHCO_3$

Risk of $NaHCO_3$ therapy

Induce **hypokalaemia** by shifting K into cells, particularly in patients with existing hypokalaemia or loss of K^+ with contracted ECF resulting normokalaemia (e.g. DM ketoacidosis)

Decrease ionic calcium level – a problem in CRF with **hypocalcemia**

Volume expansion form Na^+ load (1 mmol HCO_3^- carries 1 mmol Na^+)

- If 200 mmol $NaCO_3$ given, → 200 mmol Na^+ is given = more than 1 liter of normal saline (156 mmol Na^+)

Too rapid correction may lead to **paradoxical cerebral acidosis**

Paradoxical Cerebral Acidosis

plasma

CSF

Add NaHCO3

$\uparrow CO_2 \dashrightarrow CO_2 \uparrow$

$HCO_3^- + H^+ \rightarrow H_2O + CO_2$

Carbonic Anhydrase

$CO_2 + H_2O \dashrightarrow HCO_3^- + H^+$

HCO_3^-

$[H^+] \uparrow$

$[HCO_3] \uparrow$

BBB

Urgent IV NaHCO3 replacement

Estimation of HCO_3^- needed

HCO_3^- deficit = HCO_3^- deficit in litre x HCO_3^- space (=BW x 0.6)

usually only half the amount is given (ie.BW x 0.3)

Only replace [HCO_3^-]to a safe level acutely (pH ~ 7.1), then followed by slower correction or correction by other means

Give half of the dose initially, recheck afterwards

Beware of fluid overload, esp in oliguric patients

Renal Tubular Acidosis

Characterized by **normal AG acidosis** with:

1. **HypoK**
   - Proximal RTA (type II) – due to loss of HCO_3^-
   - Distal RTA (type I) – due to failure of H+ excretion
   - Mixed (type III)

2. HyperK
   - Type IV RTA

Proximal RTA

- Normally, HCO_3^- are filtered in glomerulus and totally reabsorbed in proximal tubule if concentration below reaborption threshold (~25 mmol/L)

- Characterized by ↓ HCO_3^- reabsorption threshold in proximal tubules
  - → loss of HCO_3^- in urine → Low plasma HCO_3^-
  - → normal AG metabolic acidosis with compensatory hyperchloremia
  - Alkaline urine despite acidosis, urine pH usually > 6

Proximal RTA

Loss of Na^+ coupling with loss of HCO_3^-

→ hypovolemia

→ hyperaldosteronism

→ hypoK^+

Associated with hyperphosphaturia, hypercitraturia (preventing nephrocalcinosis/ stones), hyperuricuria

Hyperphosphaturia → rickets, osteomalacia

Fanconi's syndrome – a pan-dysfunction of proximal tubules with amino-aciduria, glycosuria on top of RTA, hyperphosphatemia

Distal RTA

- Due to inability to excrete H^+ :
  - Failure of pumping H^+ against concentration gradient
    - H^+ATPase pump defect
  - H^+ back leak
    - Increased H+ permeability

- Urine pH always > 6 because of failure to maintain a steep plasma-urine H^+ gradient

- ↓ H^+ excretion → ↑ K^+ excretion for exchange of Na^+ reabsorption in distal tubule → hypoK^+

- Acidosis → ↑ Ca reabsorption from bone and ↓tubular Ca, PO4 reabsorption → hypercalciuria, nephrocalcinosis / stones

Causes of proximal and distal RTA

Proximal RTA (Type 2)	Distal RTA (Type 1)
Hereditary	**Hereditary**
▪ Cystinosis,	▪ Primary hypercalciuria,
▪ Galactosaemia	▪ Marfan's syndrome
▪ Wilson's disease	▪ Ehlers-Danlos syndrome
▪ Lowe's syndrome	
Acquired	**Acquired**
▪ Dysparaproteinemia	▪ Autoimmune disease: Sjogren's, RA, SLE, PBC
▪ Toxins: Heavy metal poisoning	▪ Drugs – amphotericin B, lithium, analgesic nephropathy
▪ Drugs: Carbonic anhydrase inhibitors	▪ Renal disease –chronic renal failure, urinary tract obstruction, interstitial nephritis, medullary sponge disease
▪ Renal disease: amyloidosis, renal transplant rejection, Sjogren's syndrome	▪ Paraproteinaemia, hypergammaglobulinaemia
▪ HyperPTH, hyperCa2+	▪ hyperPTH, hypervitaminD

Renal Tubular Acidosis - Diagnosis

Clinical Suspicion when:

- Normal AG acidosis with hypokalaemia
  - Hint: ↑Cl with normal $[Na^+]$

- Urine pH >5.5 in the presence of acidaemia

Confirmatory tests:

- Fractional excretion of bicarbonate (FE_{HCO3^-}): Proximal RTA
- Acid loading test (NH_4Cl): Distal RTA

Fractional Excretion of HCO3 (FE_{HCO3^-})

To test whether there is excessive loss of HCO_3^- in urine

$$FE_{HCO3^-} = \frac{\text{urine } [HCO_3^-] \times \text{plasma } [Cr]}{\text{plasma } [HCO_3^-] \times \text{urine } [Cr]}$$

Proximal RTA : > 15%

- Sensitivity increases after NaHCO3 infusion at 0.5-1.0 mmol/kg/hour to bring up the plasma HCO_3^- level, urine pH>7.5

Distal RTA: normal, i.e. <5%

NH_4Cl loading test

To test whether urine can be acidified, i.e. whether a steep H^+ gradient across plasma and urine can be maintained

Oral NH_4Cl 0.1 g/kg to bring acidosis

Normal: urine pH <5.5

Distal RTA: urine pH remains >6.0

Proximal RTA: variable

Type IV RTA

- Characterized by
  - Normal AG metabolic acidosis, and
  - hyper K^+
- Mechanism:
  - Aldosterone deficiency / resistance
- Aldosterone promotes distal K^+ and H^+ secretion, Na^+ reabsorption
  - Retention of K^+ → hyperkalaemia
  - Decreased H^+ excretion → acidosis (usually mild)
  - Urine pH variable

Type IV RTA - causes

Drugs

- ACEI, ARB
- K^+ sparing diuretics
  - spironolactone, amiloride, triamterene
- Cyclosporin A, Tacrolimus

Hyporeninism

- DMN

Renal failure

Mineralocorticoids deficiency

Kidney transplant rejection - tubulitis

Management of RTA

- **Type I and II RTA**:

  1. Correct acidosis by oral NaHCO3
     Very high dose is required in proximal RTA because of loss of HCO_3^- in urine
     Potassium citrate is a better alternative for distal RTA (citrate → HCO_3^- by liver)

  2. Potassium supplement for hypoK^+

  3. Distal RTA due to Sjogren: Steroids

- **Type IV RTA**:

  1.Stop or ↓ inciting drugs

  1.Loop diuretics + low K diet for hyperK^+ in Type IV RTA

Metabolic alkalosis

- **Causes:**
  - **Loss of H^+**
    - GI loss – **vomiting**, nasogastric drainage,
    - Renal loss
      - **Diuretics (thiazide, furosemide), hypoK**
      - mineralocorticoid excess – primary or secondary
      - Bartter's / Gitelman's syndrome
  - **Retention of HCO_3^-**
    - Intake of $NaHCO_3$
    - Milk-alkali syndrome

- **Treatment:**
  - If ECF contracted, expand with saline, HCO_3^- will fall with expansion
  - Correct hypoK^+
  - If ECF expanded, correct alkalosis with IV HCl or oral NH_4Cl

Na homeostasis

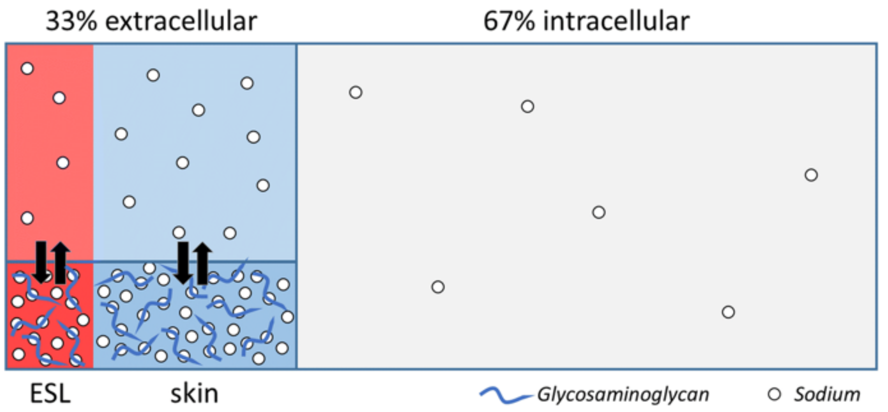
[Na^+] does not reflect absolute content of Na^+ in body, but the amount of solvent – water

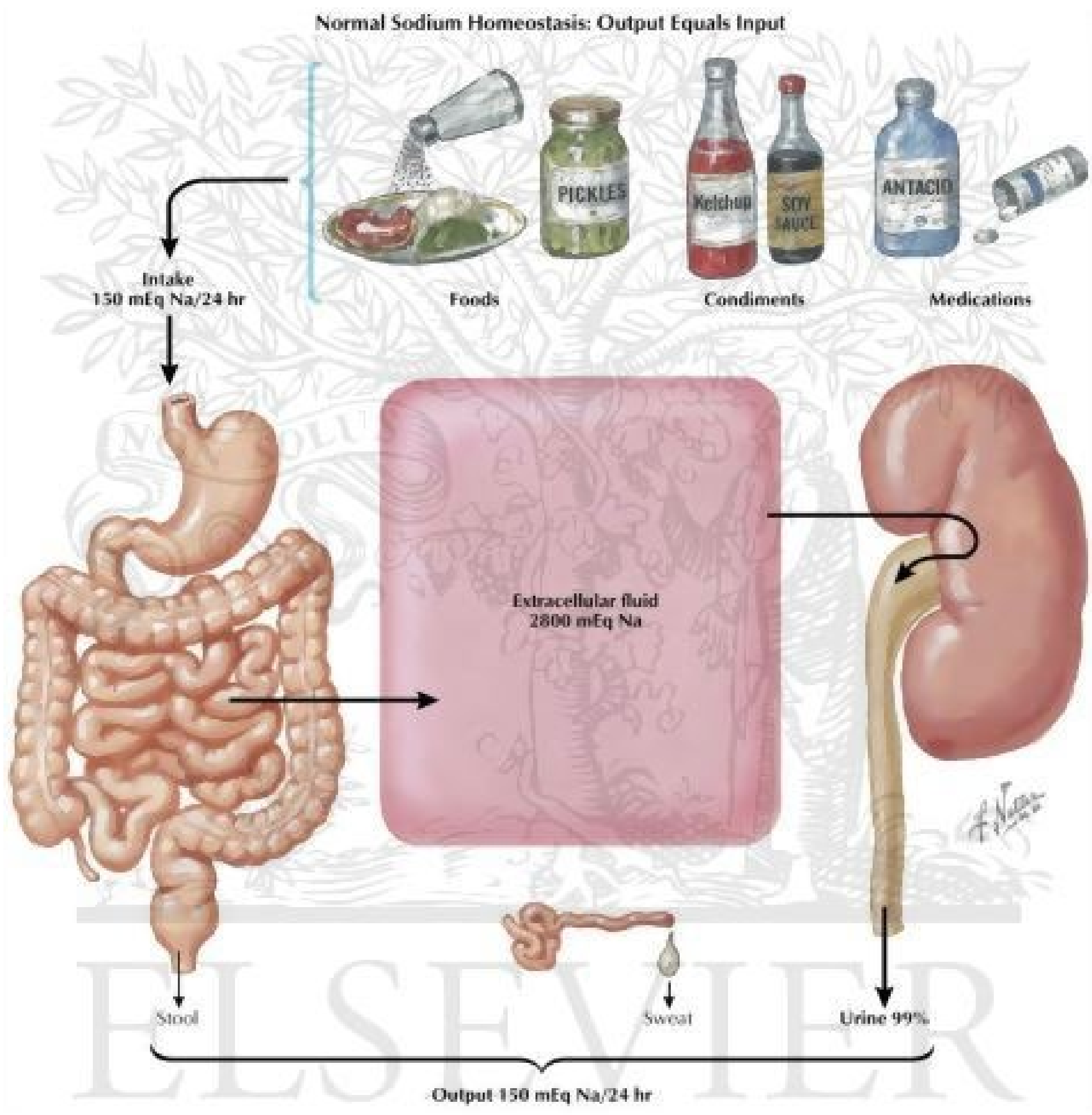
Na is primarily an extra-cellular cation

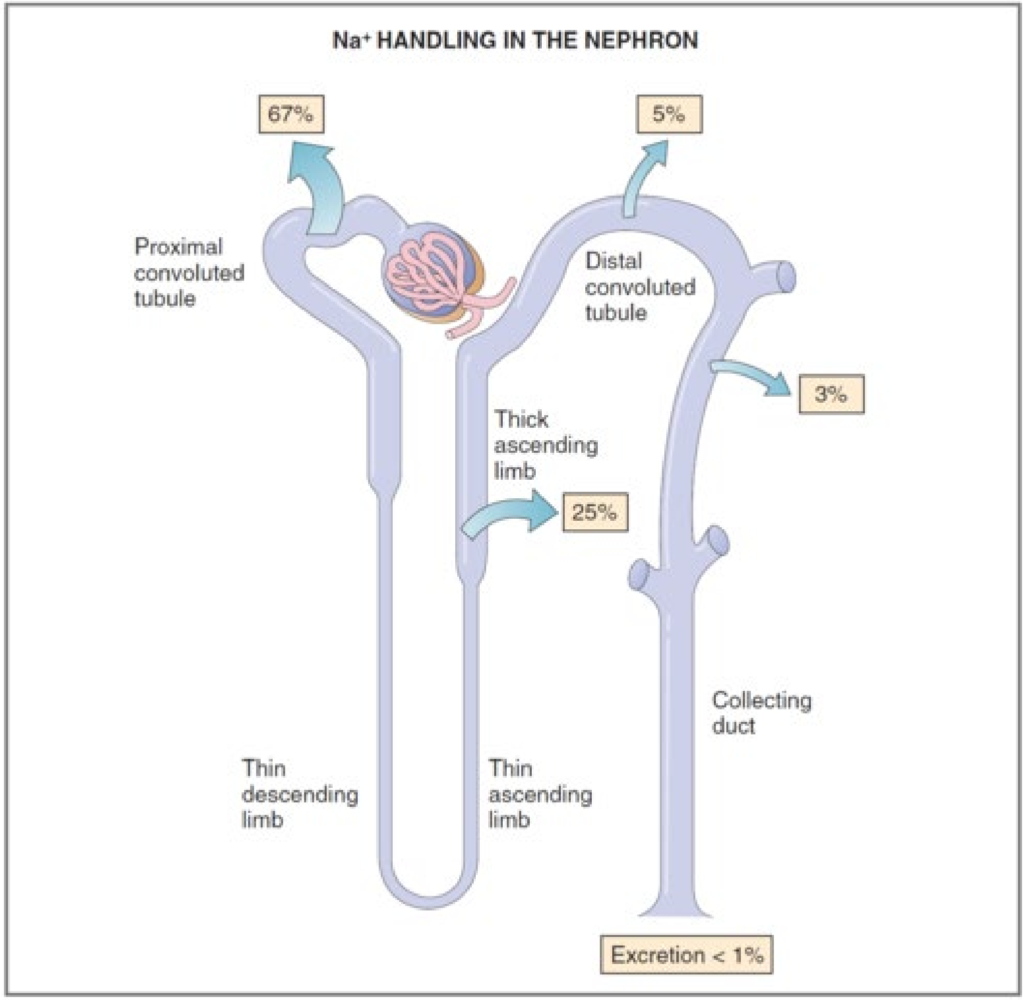
In the absence of Na loss or retention:

- hypoNa^+ => water retention
- hyperNa^+ => water depletion

Situation will be complicated when there is also Na^+ loss or retention




Normal Sodium Homeostasis: Output Equals Input
PICKLES
Ketchup
SOY SAUCE
ANTACID
Intake
150 mEq Na/24 hr
Foods
Condiments
Medications
Extracellular fluid
2800 mEq Na
Stool
Sweat
Urine 99%
Output 150 mEq Na/24 hr
ELSEVIER


Na+ HANDLING IN THE NEPHRON
67%
5%
Proximal
convoluted
tubule
Distal
convoluted
tubule
3%
Thick
ascending
limb
25%
Collecting
duct
Thin
descending
limb
Thin
ascending
limb
Excretion < 1%

Hyponatraemia

[Na^+] does not reflect absolute content of Na^+ in body, but the amount of solvent – water

In the absence of Na loss or retention:

- hypoNa^+ => water retention
- hyperNa^+ => water depletion

Situation will be complicated when there is also Na^+ loss or retention

Pseudohyponatremia

- Refer to ↓ serum [Na^+] but normal osmolality
- Due to occupation of large amount of non-water (e.g. fat, paraprotein) in plasma → ↑ plasma volume (denominator) while the actual plasma water Na^+ (nominator) is normal → [Na^+] appears to be low
- Clue: check serum osmolality – normal
- Confirmation : check plasma water [Na^+]
- Causes: Hyperglycemia, ↑↑TG paraproteinemia, ↑↑WCC

Approach to HypoNa

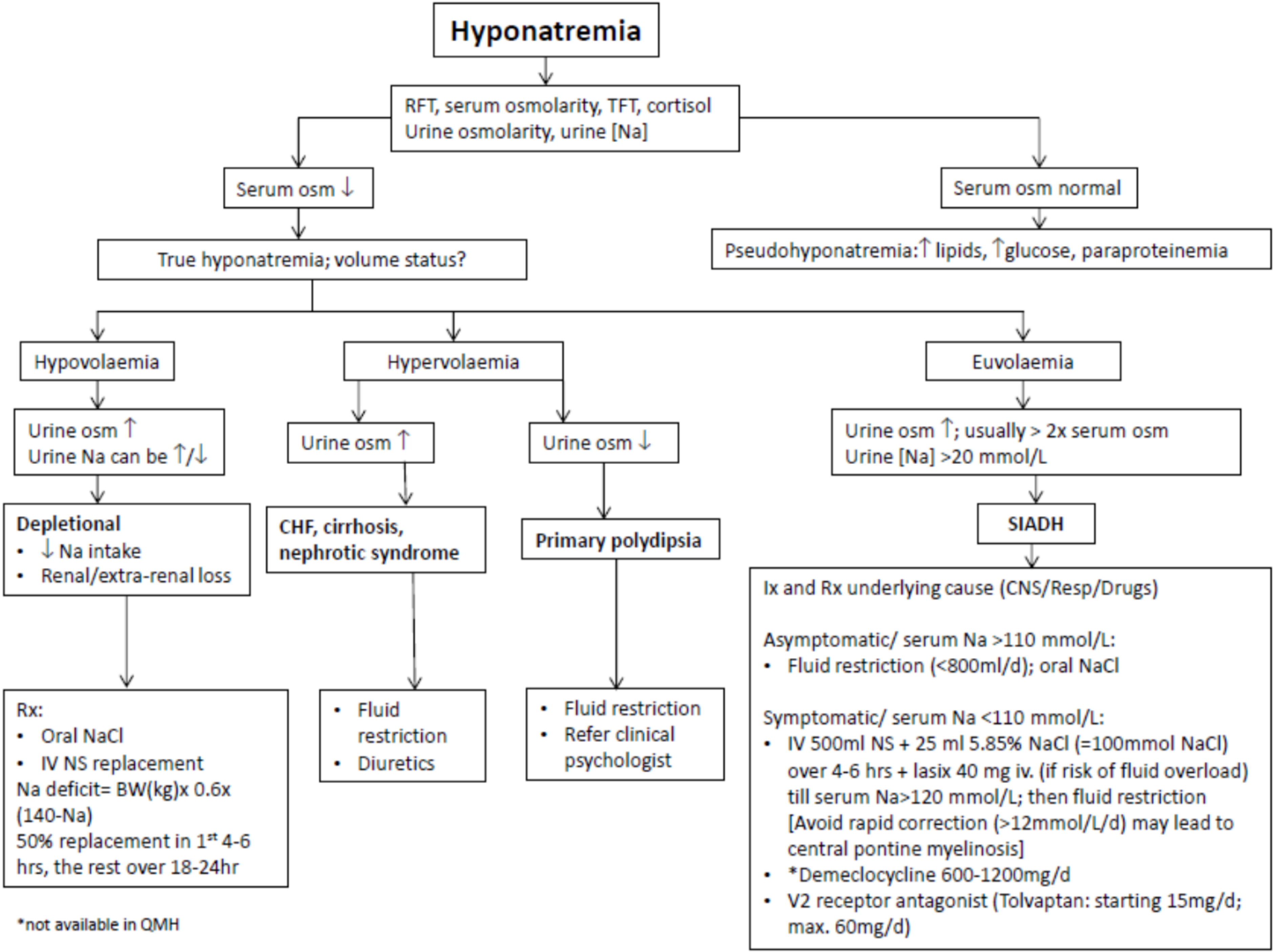
Steps in approaching HypoNa:

1. Serum osmolarity? (↓ = true hyponatremia)

2. Volume status?

3. Urine osmolarity? (>100 = ADH is working; appropriate?)

4. Urine Na level? (>20 = inappropriate ie. SIADH)


Hyponatremia
RFT, serum osmolarity, TFT, cortisol
Urine osmolarity, urine [Na]
Serum osm ↓
Serum osm normal
Pseudohyponatremia:↑ lipids, ↑glucose, paraproteinemia
True hyponatremia; volume status?
Hypovolaemia
Hypervolaemia
Euvolaemia
Urine osm ↑
Urine Na can be ↑/↓
Urine osm ↑
Urine osm ↓
Urine osm ↑; usually > 2x serum osm
Urine [Na] >20 mmol/L
Depletional
• ↓ Na intake
• Renal/extra-renal loss
CHF, cirrhosis, nephrotic syndrome
Primary polydipsia
SIADH
Ix and Rx underlying cause (CNS/Resp/Drugs)
Asymptomatic/ serum Na >110 mmol/L:
• Fluid restriction (<800ml/d); oral NaCl
Symptomatic/ serum Na <110 mmol/L:
• IV 500ml NS + 25 ml 5.85% NaCl (=100mmol NaCl) over 4-6 hrs + lasix 40 mg iv. (if risk of fluid overload) till serum Na>120 mmol/L; then fluid restriction [Avoid rapid correction (>12mmol/L/d) may lead to central pontine myelinosis]
• *Demeclocycline 600-1200mg/d
• V2 receptor antagonist (Tolvaptan: starting 15mg/d; max. 60mg/d)
Rx:
• Oral NaCl
• IV NS replacement
Na deficit= BW(kg)x 0.6x (140-Na)
50% replacement in 1st 4-6 hrs, the rest over 18-24hr
• Fluid restriction
• Diuretics
• Fluid restriction
• Refer clinical psychologist
*not available in QMH

Causes of SIADH

CNS

- Meningitis, encephalitis, brain abscess
- Head trauma, SAH, CVA, raised ICP

Respiratory

- CA lung
- Chest infection, positive pressure breathing

Drugs (e.g. SSRI, ecstasy)

Hypothyroidism

HypoNa - symptoms

- Non-specific (e.g. malaise, lethargy, headache)
- Confusion, convulsion, coma
- More serious with acute hyponatraemia

Acute vs. Chronic hypoNa

- Acute:
  - <48 hours
  - more serious symptoms, severe symptoms at $[Na^+] < 120$
  - Less complications if corrected rapidly

- Chronic:
  - >72 hours
  - Symptoms may not develop even $[Na^+] < 110$
  - More prone to CNS complications with rapid correction

Management of hypoNa

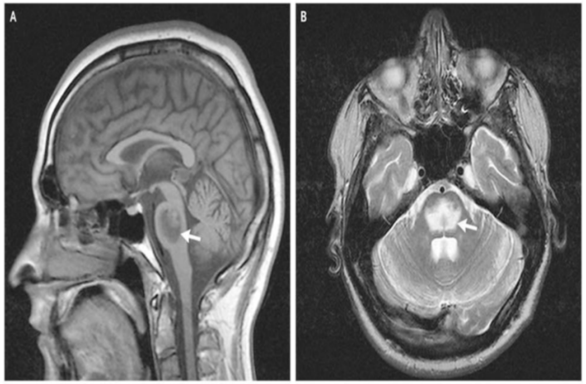
Rate of correction should be **SLOW** for chronic hyponatraemia (>2 days): <0.5 mmol/L/hr, or <12 mmol/L/day

Too rapid correction : **central pontine myelolysis**

Treat according to ECF volume status, estimate the amount of water and sodium needed

Hypertonic saline should only be used in very experienced hands!

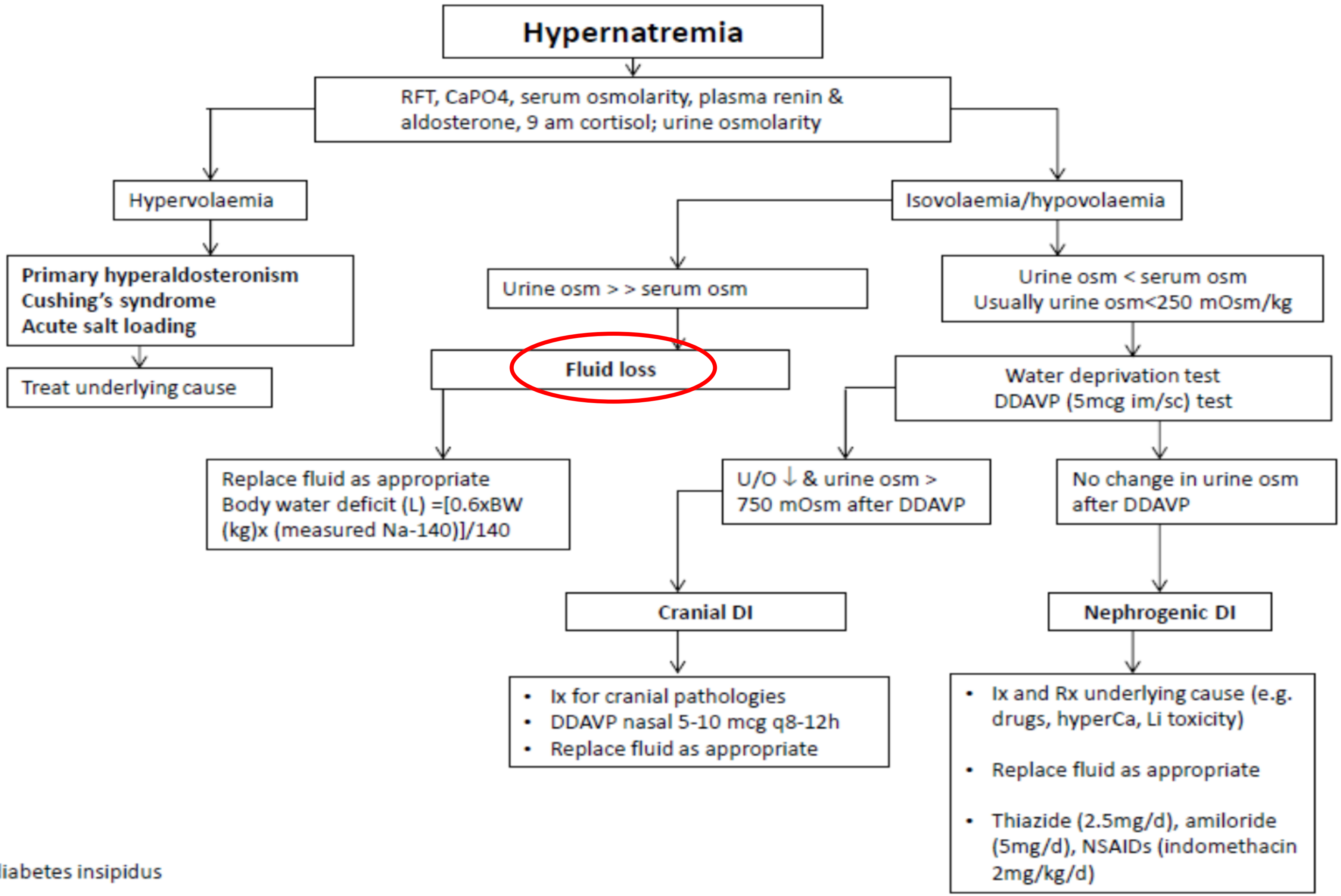
Demeclocycline or V2 antagonist can be considered in SIADH



H_2O and Na^+ replacement in contracted ECF

- Assess the degree of volume depletion or excess:
  - Mildly dehydrated: loss of skin turgor : 5% BW loss
  - Moderate: postural hypotension: 10% loss
  - Severe: shock: 15% loss
  - Mild oedema: 5% in excess

- Replace the volume depletion with normal saline cautiously
  - Initial 1/3 can be given in a first 8 hours, reduce speed afterwards
  - Monitor the [Na^+] regularly



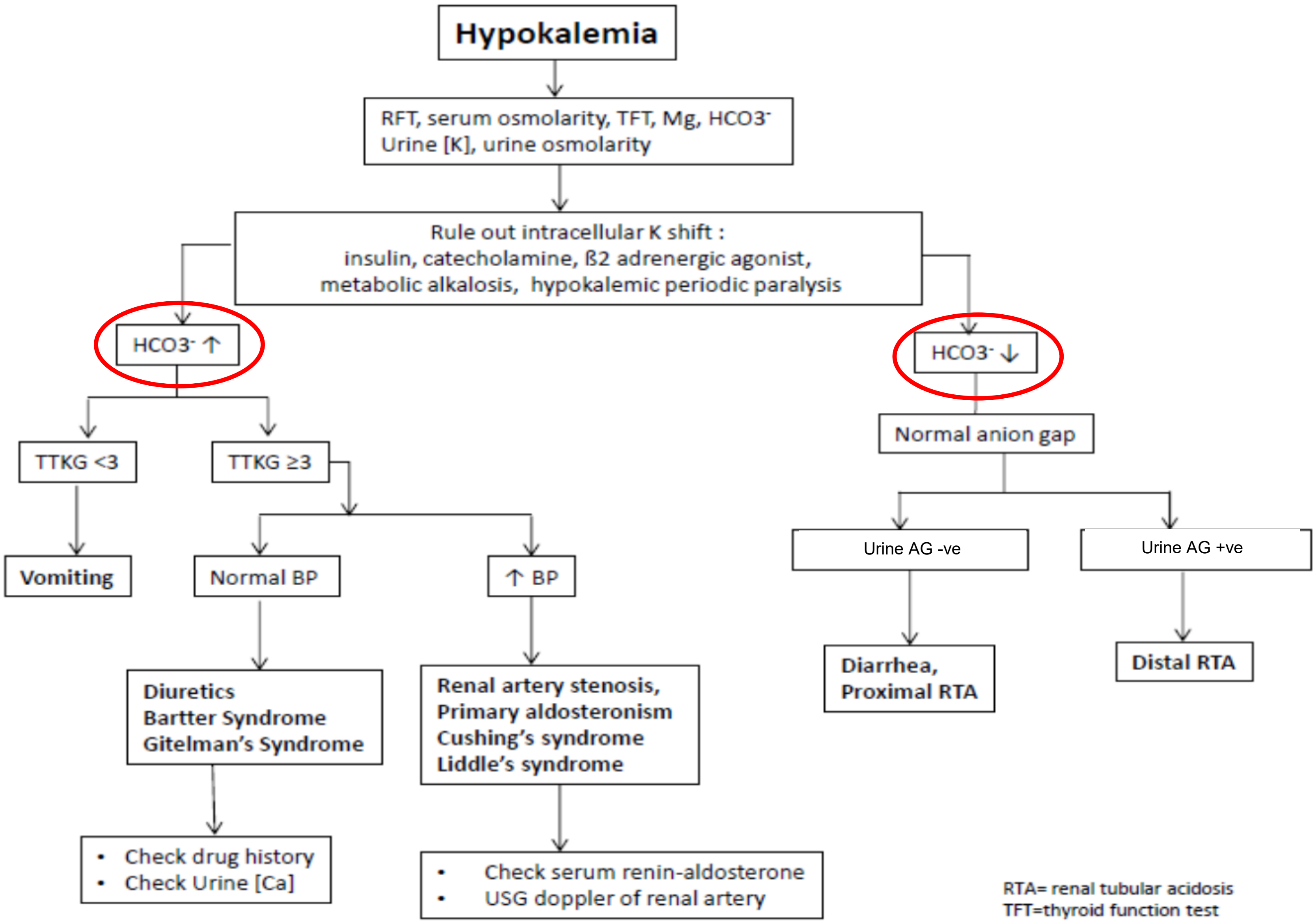
DI= diabetes insipidus

Management of HyperNa

- In pure water loss
  - Water deficit = TBW x ([Na^+]/140 - 1)

- Rate of volume replacement
  - 1/3 rule: 1/3 in first 8 hours,
    1/3 in second 16 hours,
    1/3 in third 24 hours
  - Remember to replace any ongoing water loss

- Rate of [Na^+] correction
  - Too rapid will lead to **cerebral oedema** due to rapidly reduced plasma osmolality → influx of water into cells
  - Maximal decrease of 12 mmol/24 hours unless seriously symptomatic



RTA= renal tubular acidosis
TFT=thyroid function test
TTKG=transtubular potassium gradient

K homeostasis

- K is an intracellular cation (K diffuse slowly outwards; Na-K ATPase pumps K into cells)
- Normal serum K 3.5-5.0 mmol/L (intracellular: 150 mmol/L)
- Highly influenced by acid-base status (acidosis → K moves out of the cells and vice versa)
- Severe hypoK & hyperK are life-threatening electrolyte disturbances → cardiac arrythmia

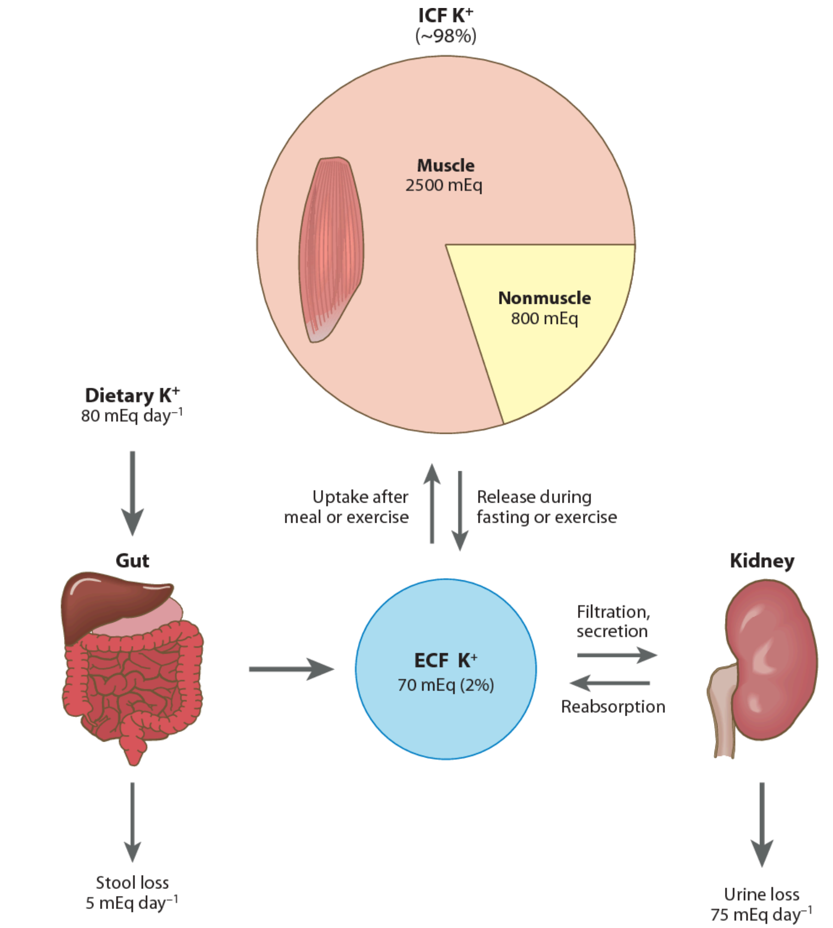


Figure 1

TTKG

Transtubular potassium gradient (TTKG):

An attempt to over-ride the interference from urinary concentration to determine the urinary K loss

$$TTKG = \frac{\text{Tubular [K]}}{\text{Serum [K]}}$$

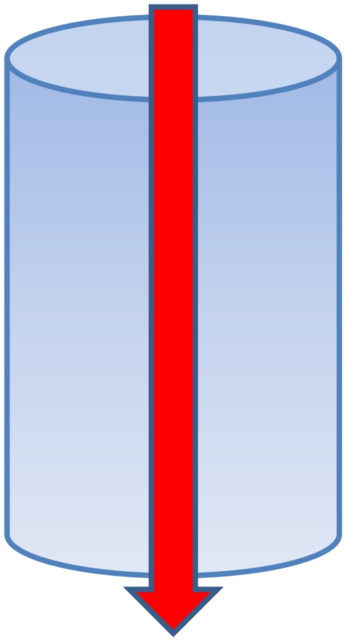
$$\text{i.e. } \frac{\text{Urine [K] x serum osm}}{\text{Serum [K] x urine osm}}$$

In the presence of hypoK, TTKG >4 : high [K^+] in terminal cortical collecting duct (i.e. renal loss)

TTKG

Serum K = 2.8 mmol/L

Serum osm = 285



Tubular K = 20

Urine osm: 570

Urine [K]: 40 mmol/L

$$\text{TTKG} = \frac{\text{Tubular [K]}}{\text{Serum [K]}}$$

$$= \frac{\{\text{Urine [K]} \times \text{serum osm/urine osm}\}}{\text{Serum K}}$$

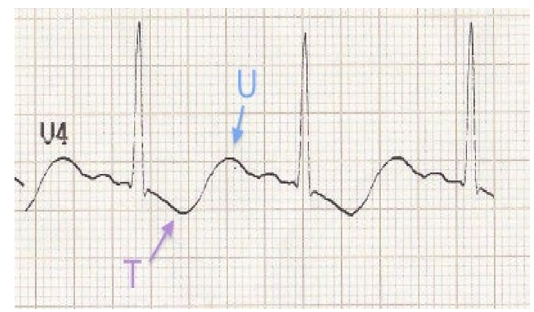
$$= \frac{\text{Urine [K]} \times \text{serum osm}}{\text{Serum [K]} \times \text{urine osm}}$$

$$= \frac{40 \times 285}{2.8 \times 570} = \mathbf{7.1}$$

Hypokalaemia – risk / complications & symptoms

Muscle weakness, paralysis (proximal muscle myopathy)

Cardiac arrhythmia, particularly when $[K^+]<2.0$



ECG changes – large U wave, loss of T wave, prolonged QT interval

Ileus, constipation

Rhabdomyolysis

Polyuria

Management of hypoK

Oral replacement when mild

Use IV KCl when:

- Oral not possible e.g. vomiting
- Moderate to severe hypoK+ (e.g. serum K $\leq$2.5)
- Always remember to dilute iv KCL in NS

Exact amount of K^+ needed cannot be calculated because

- Largely intracellular
- Discrepancy between ICF and ECF [K^+]
- Initial distribution in ECF only – 40 mmol K^+ will increase [K^+] 1.5 mmol/L

Hyperkalemia

- Exclude pseudohyperkalemia (e.g. hemolysis)
  & drugs (e.g. K supplement, NSAID, ACEI/ARB/K-sparing diuretics, salt substitute)
- Check RFT, CaPO4, HCO_3^- , ECG

Renal failure

Hypoaldosteronism
- Type IV RTA
- Addison's disease
- Congenital adrenal hyperplasia

Shift from cells
(e.g. metabolic acidosis, insulin deficiency, cell lysis)

Rx:

<u>Acute Hyperkalemia (K> 6.0mmol/l)</u>

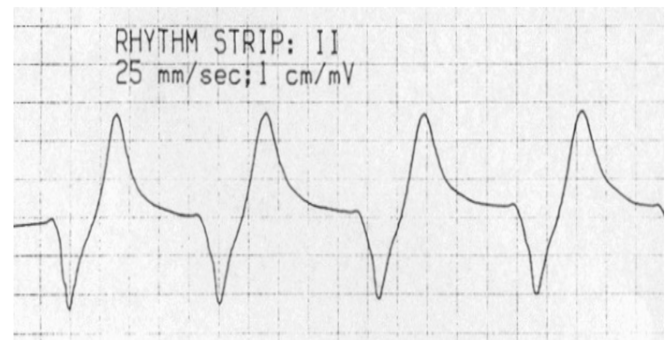
1. *Ca gluconate* (10%) 10-30ml over 5 min
2. *Glucose infusion + insulin*
   250 ml D10 or 50 ml D50 + 8 units Actrapid over 30 min
3. *NaHCO3*(8.4%) 100-150 ml over 1 h
   Useful in the presence of metabolic acidosis
   Never administered together with Ca infusion & watch out for fluid overload
4. *Frusemide*
5. *Cation-exchange resins* e.g. Resonium C Rectal: 50g; PO: 15-30 g Q6h
6. *Dialysis* if other indications for dialysis present e.g. marked acidosis, fluid overload, etc.

<u>Chronic Hyperkalaemia (K< 6.0 mmol/l)</u>

1. low K diet
2. oral frusemide
3. correct acidosis
4. Fludrocortisone 0.1-0.2mg for type IV RTA

HyperK –symptoms and risks

- Symptoms – muscle weakness, usually occurs when [K^+] > 8
- Cardiac arrhythmia
  - May occur with [K^+] >6.0
  - Rate of rise of [K^+] important – more tolerable with chronic hyperK^+
  - ECG: peak T, widening of QRS, loss of P,



HyperK – Diagnosis

- Diagnosis:
  - History, diet, drug history
- Repeat [K^+] if no apparent cause found (pseudohyperK ? hemolysis)
- P/E – volume depleted?
- Renal function test
- TTKG <6 indicates an inappropriate renal response to hyperkalemia

Management of HyperK

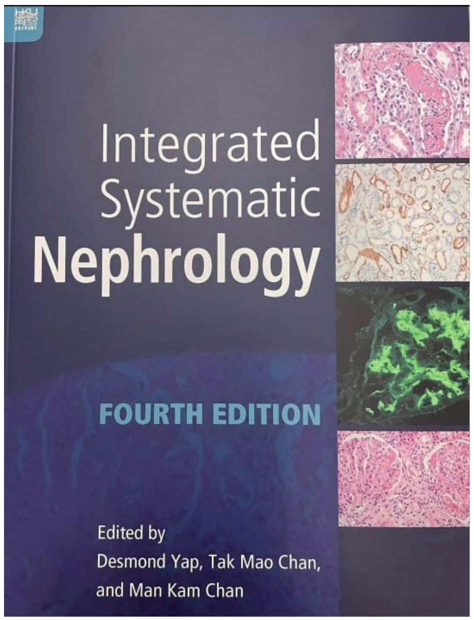
Is it an EMERGENCY? (Arrhythmia, ECG changes)

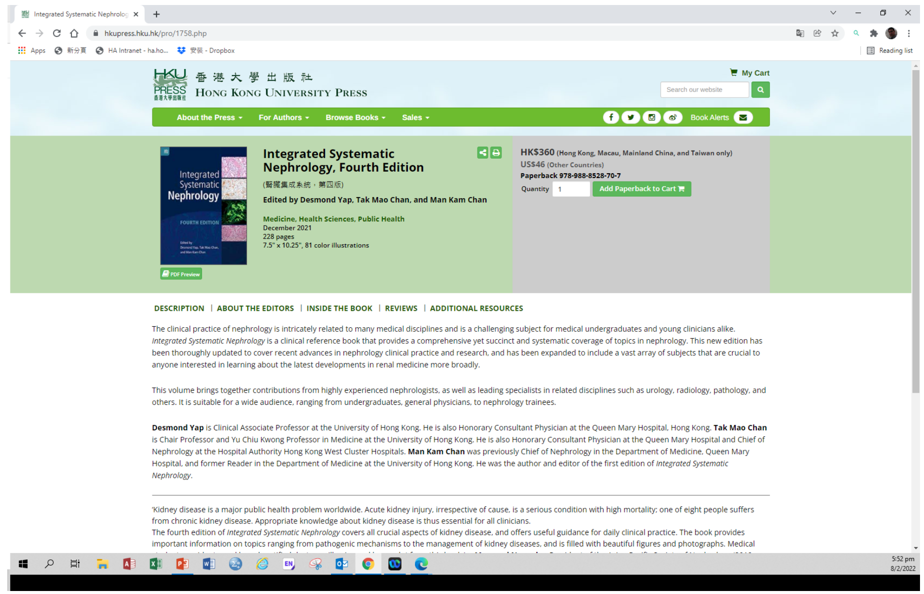
1. **IV calcium** – effect within several minutes
   Protects heart (increase threshold potential) but not reduce [K^+]
   May repeat injection 5 minutes later if effect not seen
2. **$NaHCO_3$ infusion** – shift K^+ into cells
   Effect within 30 minutes, last for several hours
3. **IV insulin/dextrose**– 10 unit to 30 gm glucose (60ml D50) – shift K^+ into cells
   Effect within 1 hour, but effect may last for only 5-6 hours
4. **Urgent HD** – most effective in removing $K^{+,}$ But takes time to set up

Non-emergency:

- Cation-exchange resin – Na/Ca resin
  - Oral or enema
- Loop Diuretics if not oliguric
- Remove underlying cause
  - Remove drugs
  - Volume expansion for depletion
- Correct metabolic acidosis
- Dialysis – HD or PD
- Low K^+ diet

香港大學出版社
HONG KONG UNIVERSITY PRESS





■THANK YOU

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!

JC
WCS

Look for today's teacher
and the others you may have missed.
(HKU Portal login required)